

# A new link prediction model for grain trade networks based on improved variational graph autoencoder and genetic algorithm

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## ABSTRACT

Food security is related to the national economy and people's livelihood, and exploring potential cooperative relationships with oneself is one of the most effective strategies to prevent and mitigate the risk of food import supply chain disruptions. How to use prior information in the grain import trade network to obtain better potential representations of nodes is a key issue in link prediction tasks. Under the theoretical framework of the variational graph autoencoder, this paper creates a new link prediction model, IVGAE-GA. Two feature extraction modules and a feature fusion module are designed to mine effective information in the trade networks. Specifically, a dynamic adaptive graph attention (DAGAN) module is proposed to extract high-order feature information from trade networks. Then, the neighborhood feature information of each node is captured through the graph convolutional neural network (GCN) to strengthen the guiding effect of the initial prior information on the prediction results. In addition, an average feature fusion (AVFF) module is designed to further refine the latent representation of nodes by mixing these non-local and local feature information. The entire IVGAE framework is optimized through cross-entropy loss and KL loss. Finally, the genetic algorithm (GA) is utilized for hyperparameter selection to help the model perform better. Extensive experimental results on two widely used publicly available datasets and four real grain trade networks illustrate that our model achieves better prediction performance compared to some existing methods. The proposed link prediction framework can be a good option for predicting potential cooperative relationships.

## 1. Introduction

Food security is an important cornerstone of global economic development and social stability [1]. Countries greatly differ in resource endowments, economic strength, and technological levels. Therefore, global food trade has become an important way to adjust the balance between supply and demand and ensure the stable operation of the global food market [2]. Studies have shown that the total annual global food trade can feed billions of people, and most people now live in net food-importing countries [3]. In recent years, pandemic crises, geopolitical conflicts, natural disasters, and other uncertain incidents have occurred frequently, which has led to the disruption of food supply chains, seriously affecting global food security [4–6]. When food supply chain disruptions cannot be effectively resolved, the crisis may propagate through the entire system, ultimately affecting sustainability development in regions and societies [7].

Rice, maize, wheat, and soybeans are the four main food crops of the world, and their trade networks are critical frameworks for stabilizing the import supply chain and ensuring global food security [8]. Many existing works have studied the structure, dynamics, vulnerability to trade

disruptions, and possible cascading effects of these grain trade networks based on network models. Based on the bilateral network of agricultural products and food-producing countries, Campi et al. [9] studied the impact of specialization patterns on food supply, food security, and food system sustainability. Taking the Pearl River Basin in China as an example, Zhou et al. [10] constructed a grain social-ecological network. The evolution of the network shows that the network model can effectively and intuitively quantify the supply and demand flows of the grain social-ecological network and facilitate food decision-making. Complex networks have been widely used to learn the robustness of grain trade networks from multiple perspectives [11,12].

In general, no matter what international trade approach a country adopts, its food supply faces many risks. As an important global issue, various researchers are seeking ways to avoid or mitigate the food crises posed by these disruption risks. Scheibe et al. pointed out that supply chain risk may be increased when overreliance on fewer partners, so establishing grain partnerships with more countries is one of the most effective strategies to mitigate these disruption risks [13].

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Alam et al. further proposed that the strength of a supply network is generally determined by its suppliers, and having diverse suppliers can improve the strength of the supply chain [6]. Deteix et al. also came to a similar conclusion through supply risk analysis, that is, trade diversification can reduce the supply risk of wheat and corn in many countries [14]. Therefore, diversification import strategies can generally be understood as one of the most effective insurance measures against supply chain disruption risk. Having multiple import suppliers enhances food security and reduces vulnerability of the import supply chain in the event of temporary or permanent supply disruptions.

Compared with traditional methods such as the gravity model [15], and similarity indicator [16], AI algorithms can provide a better fit in the long term for predicting potential trade relationships. Among deep learning methods, graph neural networks are widely used to predict whether there are potential connection relationships between unconnected nodes in the graph by virtue of their ability to effectively capture complex relationships in the graph structure [17–19]. Exploring high-precision link prediction models can identify countries or regions that have not yet established direct trade relationships but have potential trade possibilities, which is of significant value to policymakers in developing strategies to enhance the resilience of grain supply chains.

Many recent works are dedicated to leveraging graph neural networks for link prediction in trade networks [20–22]. The grain import trade networks have characteristics such as sparseness, degree imbalance, and the existence of isolated nodes (whose degree is zero). As shown in Fig. 1, trade activities between core countries are frequent, and some countries still have not joined the global soybean import trade network. Fig. 2 shows the details of importing the four grains by 266 countries from the world. From the degree distribution of each node, it can be seen that only some countries have established trade partnerships with multiple countries, and most countries have import supply chain disruption risks. Compared with other grains, soybeans have a strong import dependence risk [23].

Establishing the link prediction model must rely on grain import trade networks, and the rich characteristic information of each country is generally hidden in the complex graph structures. Most existing link prediction models assume that the influence from neighboring nodes exists in the form of constant weights or fixed constraints. However, this fixed graph structure has inherent limitations in the feature extraction process, node features tend to become more similar with the increase of the graph neural network depth (i.e., the over-smoothing problem [24]). The degree imbalance characteristic will aggravate this problem because the features of high-degree nodes lose some personalized information after aggregating too many neighbor features. Ahn et al. [25] verified that the features of isolated nodes tend to be close to zero after the feature extraction process, which makes these isolated nodes also do not work well in link predictions. In addition, the choice of hyperparameters has a crucial impact on the performance of link prediction models. Due to the complexity of the model, it is difficult to determine the best combination of hyperparameters through theoretical analysis. Therefore, it is necessary to try different hyperparameter combinations through extensive experiments to find the most suitable hyperparameters for datasets. These experiments consume a lot of computing resources and time, and it may still be difficult to find the best hyperparameter combination.

To solve these problems effectively, this paper develops a novel link prediction model based on improved variational graph autoencoders and the genetic algorithm, referred to IVGAE-GA. The main idea is to mine prior information in the grain import trade network and extract the characteristics of nodes from both non-local and local aspects based on the edge weight matrix and adjacency matrix, respectively, to make accurate predictions. We first designed a dynamic adaptive graph attention (DAGAN) module based on edge weight matrix to capture the high-order non-local features of nodes. The technology of randomly dropping edges (DropEdge) not only breaks the limitations of the initial fixed graph structure by reconstructing the graph structure,

but also further perturbs attention coefficients. The adaptive feature extraction method is used to make the aggregation method between initial features and extracted features more sensible. Under the theory of variational graph autoencoder, the  $\ell_2$ -norm is used to normalize the features of isolated nodes so that they can obtain more feature information. Simultaneously, a local feature extraction module based on the graph convolutional neural network (GCN) is created to mine the initial prior information by capturing the characteristics of neighbor nodes. Furthermore, non-local and local information of nodes is hybrid in the form of average feature fusion (AVFF), aiming to refine the latent representation of nodes. The overall framework is optimized through cross-entropy loss and KL loss. Finally, we use the genetic algorithm (GA) to optimize the hyperparameters of the model and simplify the entire hyperparameter selection process in the form of intelligent algorithms, which not only saves time costs but also obtains better prediction results.

The main contributions of this paper are summarized as follows.

- A new end-to-end deep graph neural network is constructed for link prediction, effectively solving the fixed constraints of the initial graph structure, the zero-norm tendency of isolated nodes, and the over-smoothing problem, etc. The application of this model provides a novel research perspective for countries in food supply chain networks to predict their potential trade relationships, which can be used to mitigate the disruption risk of trade networks.
- Based on the edge weight matrix and adjacency matrix, two branches, non-local and local, are constructed respectively, which can fully mine the prior information and graph structure information from the real grain import trade network.
- Based on DropEdge, adaptive aggregation, and  $\ell_2$ -norm technology, a DAGAN module is designed that can efficiently capture high-order features in the graph structure, which effectively alleviates the inherent over-smoothing problem of deep graph neural networks.
- The genetic algorithm is utilized to intelligently tune six key hyperparameters to improve model performance and save selection costs, thereby achieving an end-to-end link prediction model.

In the rest of this paper, some preliminaries are briefly introduced in Section 2. The details of the proposed method are presented in Section 3. The experiment results and discussion are elaborated in Section 4, followed by the conclusion in Section 5.

## 2. Preliminaries

### 2.1. Notation

Generally, the grain import trade network uses the national entities participating in the transaction as network “nodes”, and the trade relationships between countries are abstracted into “edges” between nodes. Depending on the strength of the supply and demand relationship, the direction and weight of the connecting edges vary. More formally, grain import trade networks are directed graphs, and can be expressed as  $G = (V, E, A, T)$ , where  $V = \{1, \dots, m\}$  and  $E \subseteq V \times V$  are vertex set and edge set respectively. Adjacency matrix  $A = (A_{ij}) \in \mathbb{R}^{n \times n}$  is used to constrain the structure of trade networks, where  $A_{ij} = 1$  if there is an edge between country  $i$  and country  $j$ . The edge weight matrix  $T$  describes trade flows between countries.

### 2.2. Variational graph AutoEncoder

In recent years, graph autoencoders (GAE) [26] have attracted considerable attention in link prediction tasks due to their strong ability to learn efficient graph representations. As a variant of GAE, the variational graph autoencoder (VGAE) effectively handles uncertainty and

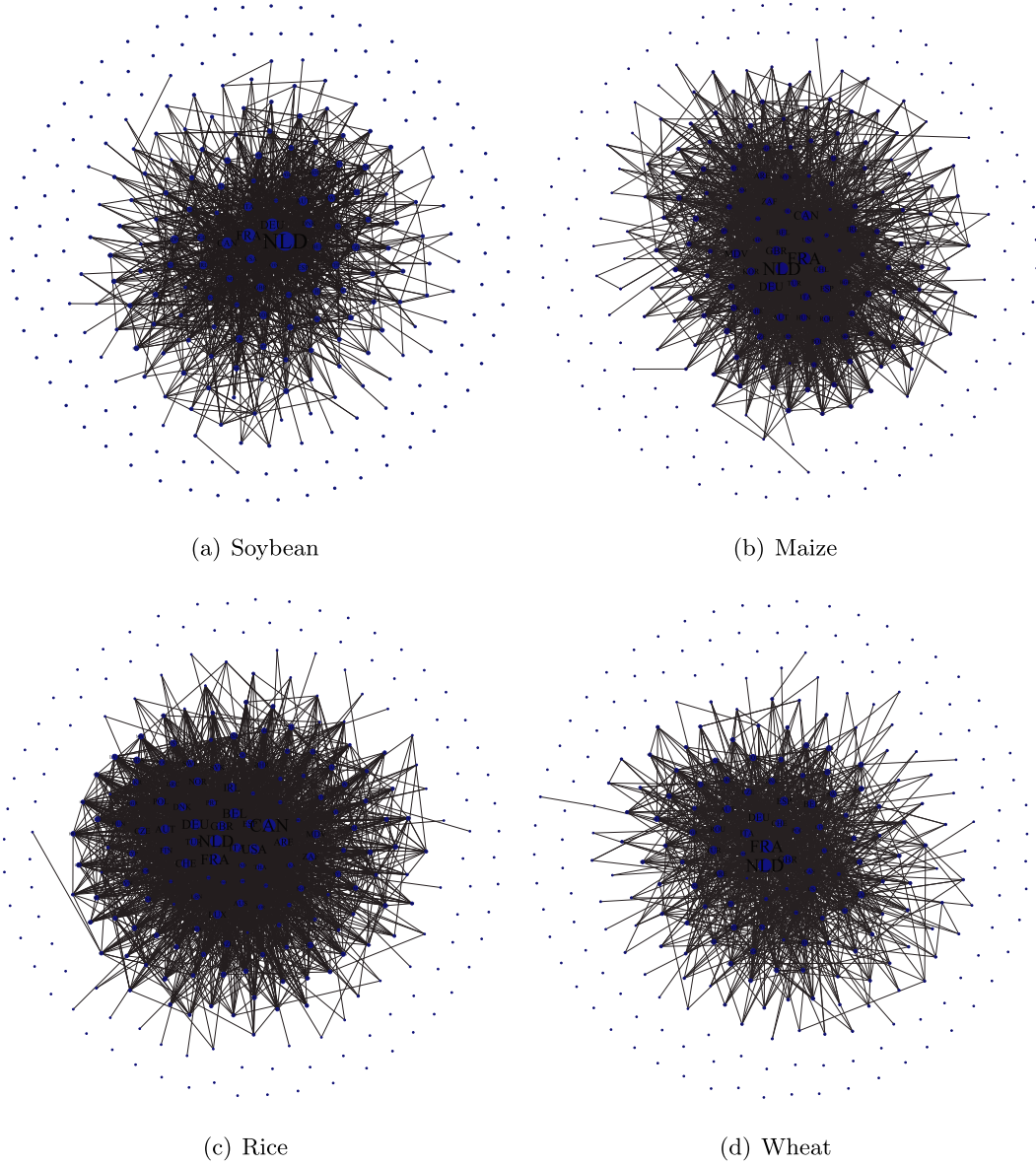


Fig. 1. Overview of global grain import trade networks in 2022, including countries, relationships, and trade flows.

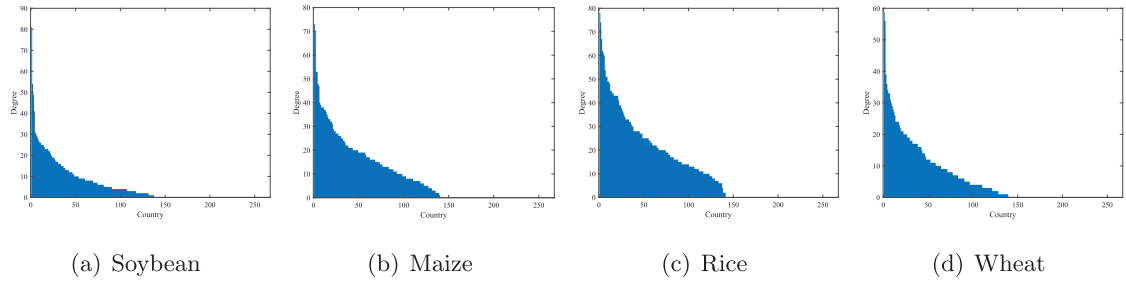


Fig. 2. Illustration of the degree of each node in grain import trade networks. There are 266 countries in the import market.

noise in graph data by learning the probability distribution of graph-structured data, thereby improving the robustness of the link prediction model. VGAE constructs inference models to map the initial feature matrix  $\mathbf{X}$  into a latent representation space based on Gaussian distribution and learns the distribution parameters of nodes in the latent representation space. Specifically, the two-layer graph convolutional neural networks are used to learn the mean  $\mu$  and variance  $\sigma$ , and the

whole process can be described as

$$\begin{aligned} \mu &= \tilde{\mathbf{A}} f(\tilde{\mathbf{A}} \mathbf{X} \mathbf{W}_0) \mathbf{W}_1, \\ \log \sigma &= \tilde{\mathbf{A}} f(\tilde{\mathbf{A}} \mathbf{X} \mathbf{W}_0) \mathbf{W}_2, \end{aligned} \quad (1)$$

where  $\mathbf{W}_i$  denote weight matrices,  $f(\cdot)$  is the rectified linear unit (ReLU) function,  $\tilde{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} = (\mathbf{D} + \mathbf{I}_n)^{-\frac{1}{2}} (\mathbf{A} + \mathbf{I}_n) (\mathbf{D} + \mathbf{I}_n)^{-\frac{1}{2}}$  is the symmetric normalization of the adjacency matrix  $\mathbf{A}$ , and  $\mathbf{D}$  is the

**Algorithm 1** Genetic algorithm is applied on the IVGAE

**Input:** Graph  $G = (V, E, A, T)$ ; feature matrix  $X$ , output channel  $c$ , learning rate  $\eta$ , number of DAGAN modules  $k$ , layer  $L$ , dropping rate  $p$ , scaling constant  $s$ .

**Step 1:** Randomly generate initial populations within a given range;

**Step 2:** Measure the fitness of the candidate population through the AP evaluation indicator;

**Step 3:** Choose the best model and keep the parameters;

**Step 4:** Blend crossover is performed on the selected parent generation to generate new candidate populations;

**Step 4:** Change certain genes of individuals through Gaussian mutation to increase the diversity of the population; go to **Step 2**. The desired output is the optimally performing hyperparameter population.

**Output:** Return the best hyperparameter population.

corresponding degree matrix of  $A$ . Therefore, the posterior distribution of the node potential representation  $Z$  generated by sampling is

$$q(Z | X, A) = \prod_{i=1}^N q(z_i | X, A), \text{ with} \quad (2)$$

$$q(z_i | X, A) = \mathcal{N}(z_i | \mu_i, \text{diag}(\sigma_i^2))$$

Then, a generative model is designed to calculate the probability of an edge existing between any two nodes to reconstruct the graph structure. The prior probability distribution of the newly generated graph structure  $A$  is:

$$p(A | Z) = \prod_{i=1}^N \prod_{j=1}^N p(A_{ij} | z_i, z_j), \text{ with } p(A_{ij} = 1 | z_i, z_j) = \sigma(z_i^\top z_j) \quad (3)$$

where  $\sigma(\cdot)$  is the logistic sigmoid function.

### 2.3. Genetic algorithm

The genetic algorithm is one of the most well-known evolutionary algorithms and has recently been extensively used for hyperparameter optimization and neural architecture search [27,28]. In fact, the genetic algorithm simulates the natural selection process of species in nature, continuously eliminating the worst-performing populations from parent to child, thereby iteratively searching for better solutions. As shown in Algorithm 1, we introduce a genetic algorithm to optimize the six hyperparameters in the proposed model automatically. Obviously, this smart search for the optimal architecture of IVGAE significantly reduces the cost of hyperparameter selection.

## 3. Methodology

In this section, we first introduce the overall framework of the proposed IVGAE-GA and then present in detail the basic idea and implementation steps for each module, including the dynamic graph attention module, the average feature fusion module, and the loss function.

### 3.1. Model framework

The whole framework of the IVGAE-GA model is shown in Fig. 3. The upper part shows the structure of our model IVGAE, and the lower part describes the process of using the genetic algorithm to explore the optimal architecture of the model. The IVGAE takes the feature matrix, edge weight matrix, and adjacency matrix as input to utilize the prior information in the real grain import trade network fully.

Two purposely designed branches in our model ensure that each node can effectively learn its non-local and local feature representations. The DAGAN modules not only solve the limitation of fixed

constraints in initial graph structures by reconstructing the graph structure multiple times but also extract rich deep feature information in the form of adaptive graph attention. This non-local feature information reveals the influence of peripheral nodes on central nodes in the trade network and plays a crucial role in discovering new edges in the link prediction task. GCN is used to capture the complex relationships between nodes and their neighbors by aggregating the neighborhood information of nodes. To ensure the effectiveness of the initial prior information, we use the average feature fusion technology to fuse the non-local and local latent features of nodes. Finally, we mix the feature representations obtained by the two branches through the sum operation to easily obtain the final latent feature of each node.

### 3.2. Dynamic adaptive graph attention module

The initial prior knowledge in the real grain import trade network is all the information we can use, so it is a natural thought to use the feature matrix  $X \in \mathbb{R}^{n \times f}$  and the adjacency matrix  $A \in \mathbb{R}^{n \times n}$  as the input of our model. To distinguish different types of grain features, we set a trainable feature matrix with the same size as the feature matrix  $X$ . Then, it is concatenated with the feature matrix according to the dimension of the row, as follows:

$$\tilde{X} = \text{Concat}(X, X'). \quad (4)$$

Existing works [29] have proven that the entanglement of representation transformation and propagation in the feature aggregation process significantly affects model performance. After decoupling these two operations, deeper graph neural networks can be used to learn graph node representations from a larger receptive field. As depicted in Fig. 4, the deep adaptive graph attention method is designed.

In the transformation operation, we use the multi-layer perceptron (MLP) to change the dimension of the feature matrix and map it to the lower dimensional space:

$$M^{(0)} = \tilde{X}W_1 + b_1, \quad (5)$$

where  $W_1$  and  $b_1$  are the weight matrix and bias respectively. To solve the “norm-zero tendency of isolated nodes” problem [25], we use  $\ell_2$ -norm to normalize the initial features matrix to constrain the feature learning process. Let the feature transformed matrix denote  $M^{(0)} = [m_1, m_2, \dots, m_n]^\top$ , and then compute the normalized matrix can be defined as

$$\tilde{m}_1 = s \frac{m_1}{\|m_1\|_2}, \quad (6)$$

$$\tilde{M}^{(0)} = [\tilde{m}_1, \tilde{m}_2, \dots, \tilde{m}_n]^\top,$$

where the  $\|\cdot\|_2$  is  $\ell_2$ -norm,  $s$  is scaling constant. As mentioned above, there are not only isolated nodes but also inherent over-smoothing problems in real grain import trade networks. To overcome the limitations of the feature extraction process caused by the fixed graph structure, we use the DropEdge [30] operation to reconstruct trade networks. This approach can not only improve the model's generalization ability through the generated adjacency matrix variants but also enhance data diversity. The edges in the initial trade network are randomly dropped according to a certain proportion, and the relationship between the reconstructed adjacency matrix  $A_{\text{drop}}$  and the initial adjacency matrix  $A$  can be expressed as:

$$A_{\text{drop}} = A - A'_p, \quad (7)$$

where  $p$  is the dropping ratio. The entire process of dropping edges is shown in Fig. 4. In international trade, trade flows between countries are crucial for predicting potential partnerships. Therefore, we utilize the mask operation to construct a novel attention matrix based on the edge weight matrix  $T$ . The process of constructing the attention matrix is described as follows.

$$\tilde{P}_{\text{drop}} = \text{softmax}(T \odot A_{\text{drop}}), \quad (8)$$



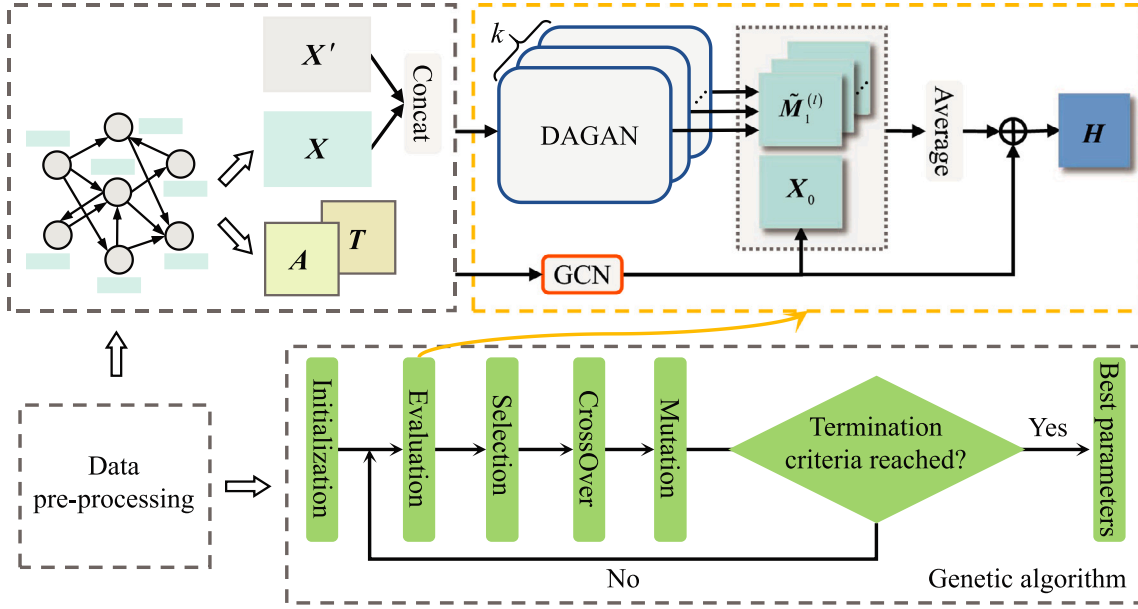


Fig. 3. The overall architecture of the proposed IVGAE-GA model.

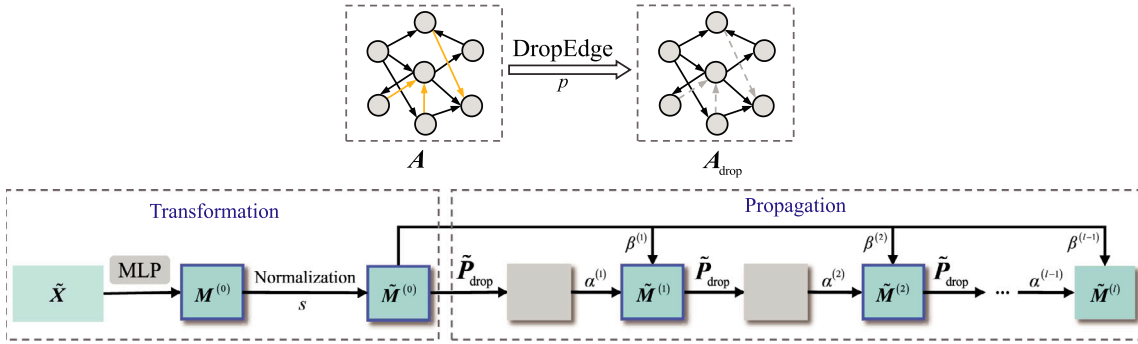


Fig. 4. Details of hybrid fusion graph attention module.

where  $\odot$  denotes the element-wise product. The softmax function is applied to normalize the attention matrix by columns.

In the propagation operation, recent works have demonstrated empirically that the performance of models is improved significantly by removing non-linear activation functions and feature transformations operation in the feature aggregation process [31,32]. In addition, the initial prior information contains some real node characteristics, and its importance is self-evident [33]. Hence, this paper defines the aggregation function as:

$$\tilde{M}^{(l)} = \alpha^{(l-1)} \tilde{P}_{drop} \tilde{M}^{(l-1)} + \beta^{(l-1)} \tilde{M}^{(0)}, \quad l = 0, 1, 2, \dots, L \quad (9)$$

where  $\alpha$  and  $\beta$  are adaptive parameters, which can balance the information from neighbors and initial features. Furthermore, we reconstructed the initial trade network  $k$  times. The feature matrix  $\tilde{M}_i^{(l)}$  ( $i = 1, 2, \dots, k$ ) can be obtained through the  $i$ th DAGAN module. Specifically, each DAGAN module can capture different feature information from different graph structures, which helps extract richer features and improve the generalization ability of our model.

### 3.3. Average feature fusion module

As shown in Fig. 3, GCN [34] is used to capture local feature information. In fact, aggregating the features of the initial node and its neighbors can effectively enhance the guiding role of the initial prior information in predicting:

$$X_0 = \text{GCN}(\tilde{X}). \quad (10)$$

Then, the average feature fusion technique to fuse the non-local and local features of nodes:

$$\begin{aligned} \tilde{M} &= \text{Stack}(X_0, \tilde{M}_1^{(l)}, \tilde{M}_2^{(l)}, \dots, \tilde{M}_k^{(l)}), \\ \hat{M} &= \text{Average}(\tilde{M}). \end{aligned} \quad (11)$$

To highlight the importance of initial prior information, we obtain the final feature matrix by summing the latent representations of nodes:

$$H = \hat{M} + X_0. \quad (12)$$

### 3.4. Loss function

As described in Section 2.2, we use the IVGAE model to learn the mean  $\mu$  under the framework of variational graph autoencoders. Specifically,  $\mu = [\mu_1, \mu_1, \dots, \mu_n]^T = \text{IVGAE}(X, A, c, \eta, k, L, p, s)$  is the matrix of mean vectors  $\mu_i$ . To avoid introducing bias, we remove the operation of normalizing the initial feature matrix in the IVGAE model. Similarly,  $\log \sigma = [\log \sigma_1, \log \sigma_2, \dots, \log \sigma_n] = \text{IVGAE}(X, A, c, \eta, k, L, p)$ . Finally, a probability distribution closer to the real characteristics of nodes is obtained.

Then, we optimize the entire model through the variational lower bound  $\mathcal{L}$ , as follows:

$$\mathcal{L} = \mathbb{E}_{q(\mathbf{Z}|\mathbf{X}, \mathbf{A})} [\log p(\mathbf{A} | \mathbf{Z})] - \text{KL}[q(\mathbf{Z} | \mathbf{X}, \mathbf{A}) \parallel p(\mathbf{Z})], \quad (13)$$

where  $\text{KL}[q(\cdot) \parallel p(\cdot)]$  represents the Kullback-Leibler divergence between the posterior distribution  $q(\cdot)$  and the prior distribution  $p(\cdot)$ . In

**Table 1**  
The statistics of the datasets.

| Dataset  | Type     | #Nodes | #Edges | #Features | Sparsity |
|----------|----------|--------|--------|-----------|----------|
| Cora     | Citation | 2708   | 5429   | 1433      | 0.14%    |
| CiteSeer | Citation | 3327   | 4732   | 3703      | 0.08%    |
| Soybean  | Trade    | 159    | 1510   | 22        | 5.97%    |
| Maize    | Trade    | 167    | 2485   | 22        | 8.91%    |
| Rice     | Trade    | 178    | 3337   | 22        | 10.53%   |
| Wheat    | Trade    | 158    | 1619   | 22        | 6.49%    |

addition, we use Gaussian prior  $p(\mathbf{Z}) = \prod_{i=1}^n \mathcal{N}(\mathbf{z}_i | \mathbf{0}, \mathbf{I})$  to constrain the proposed model.

## 4. Experiments

In this section, we conduct experiments on four real grain trade datasets and two publicly available citation datasets. Experiments using the genetic algorithm are performed on the same software configuration — an NVIDIA RTX 2080 SUPER GPU on Windows 10 and an Intel(R) Core(TM) i7-10700 CPU with 16 GB main memory.

### 4.1. Datasets

We evaluate the proposed model on various types of attributed graphs, and the statistics are summarized in Table 1. In addition to real grain import trade networks, we specially select two benchmark datasets for the link prediction task to verify the excellence of our model in this article.

#### 4.1.1. Benchmark datasets

Cora and CiteSeer datasets are widely used in link prediction tasks [26]. The Cora dataset is an academic paper dataset in the field of machine learning. It mainly contains 2708 publications, and there are 5249 citation relationships between publications. Furthermore, each publication has 1433 dimensions of features represented by 0 and 1. The CiteSeer dataset is a dataset of academic papers from the field of computer science, including 3327 papers and 4732 citation relationships. The feature dimension of each paper is 3703. Compared with grain import trade networks, these two benchmark datasets are more sparser and have larger graph structures. Therefore, the excellent performance of our model on these two data sets means that the design in this paper is rational and effective.

#### 4.1.2. Real-world grain trade networks

**Data collection.** Soybean, maize, rice, and wheat are the most important grain crops in the world, so we intentionally selected the global import data from these four crops in 2022 as research samples in link prediction tasks. To conduct our research, we extracted data from the United Nations Commodity Trade Statistics Database (<https://comtradeplus.un.org/>), which documents bilateral import and export flows across countries in the real world. In this database, we selected the harmonization system (HS) as the item classification standard and then chose soybean imports (HS 1201), maize imports (HS 1005), rice imports (HS 1006), and wheat imports (HS 1001). In addition, we collected attribute information of various countries worldwide in 2021 from the World Bank (<https://www.worldbank.org/en/home>), including 22 attributes such as population, agricultural land, GDP per capita, and land area.

**Data pre-processing.** After obtaining the raw data, we first took the data from the World Bank as the standard to standardize the names of countries and eliminated relevant trade data of countries with absent attribute information. To maintain the original distribution characteristics of the data and avoid introducing unreasonable bias, we assigned 0 to the missing values of each country. Then, the Z-score normalization technique was employed to standardize each attribute

**Table 2**  
Ablation study on different combination models on Cora dataset.

| Model                        | Framework      | AUC                 | AP                  |
|------------------------------|----------------|---------------------|---------------------|
| DAGAN+STACK                  | no-local       | 94.43 ± 1.52        | 95.15 ± 1.22        |
| DAGAN+ADFF                   | no-local       | 95.24 ± 0.24        | 95.99 ± 0.15        |
| DAGAN+AVFF                   | no-local       | 95.40 ± 0.24        | 96.00 ± 0.14        |
| [DAGAN, GCN]+AVFF            | no-local+local | 95.56 ± 0.16        | 96.10 ± 0.14        |
| [DAGAN, GCN]+AVFF+GCN (ours) | no-local+local | <b>95.66 ± 0.21</b> | <b>96.13 ± 0.14</b> |

to eliminate the impact of dimensional differences between different attributes on the model. Additionally, normalizing the attribute values to a distribution with a mean of 0 and a standard deviation of 1 allows the model to converge better. Finally, the processed data can be fed into the baseline model and our model as feature matrix.

### 4.2. Experiment setup

**Baselines.** This paper specifically selects some link prediction models based on graph autoencoders as the baseline to compare the performance of our model comprehensively. Classic methods include GAE and VGAE [26]. Then, many variants gradually emerged, including graph normalized autoEncoder (GNAE) [25], variational graph normalized autoEncoder (VGNAE) [25], multi-scale variational graph autoencoder (MSVGAE) [35], FE-GNAE [18], normalized auto-encoder based on biased walk (BW-NAE) [36], BW-VNAE [36].

**Parameter settings.** The IVGAE-GA model is trained under the PyTorch programming framework based on the Python language and optimized through the Adam [37] algorithm. Following previous link prediction models, we input the node attributes in the benchmark dataset into the model as feature matrices. For grain import trade networks, we collect 22 public attributes from the World Bank and construct a personalized feature matrix of the same size using the Xavier initializer [38]. As described above, IVGAE-GA has six hyperparameters in total: (1)  $c$ ; (2)  $\eta$ ; (3)  $k$ ; (4)  $L$ ; (5)  $p$ ; (6)  $s$ . Since there are many combinations of hyperparameters and their value range is wide, we consider using the genetic algorithm to tune all hyperparameters. In the initial experiments, we simply set  $\eta$ ,  $k$ ,  $L$ ,  $p$  and  $s$  in all data sets to [0.001, 3, 3, 0.5, 1.8]. The output channel  $c$  is related to the spatial size of the feature map, so we set it to 128, 256, 10, 10, 10, and 10 in the Cora, CiteSeer, soybean, wheat, corn, and rice datasets respectively. Referring to the evaluation methods in mainstream link prediction models, this article adopts the area under the ROC curve (AUC) and average precision (AP) scores to evaluate the performance of the proposed model. Larger AUC and AP value denotes better prediction effectiveness. We run ten repeated times and report the average and standard deviation.

### 4.3. Ablation study

To prove the excellence of IVGAE and the effectiveness of each module, we specially selected the Cora benchmark dataset to conduct ablation experiments on various combination models. Table 2 shows the predictive performance of the combination models under different architectures. In our model, the DAGAN module is an important component that can effectively capture high-order feature information and graph structure information. Therefore, we also conducted comparative experiments on the three technologies in this component, and the results are shown in Table 3.

Recall that the framework of our model can be simplified into two branches, one is the non-local feature extraction branch based on DAGAN, and the other is the local feature extraction branch based on GCN. After the process of feature extraction, we tried three methods of processing fine features, including two feature fusion methods (average feature fusion (AVFF) and adaptive feature fusion (ADFF)), and a simple feature stacking (STACK) technology. As depicted in Table 2, we combined a variety of models, and the performance of the

**Table 3**

AUC and AP test results on Cora dataset. ①: introducing initial information, ②: normalizing feature matrix., and ③: adaptive feature aggregation method.

| Model        | ① | ② | ③ | AUC                 | AP                  |
|--------------|---|---|---|---------------------|---------------------|
| A            |   | ✓ | ✓ | 72.76 ± 2.73        | 74.78 ± 1.88        |
| B            | ✓ |   | ✓ | 71.96 ± 1.37        | 74.50 ± 1.03        |
| C            | ✓ | ✓ |   | 95.27 ± 0.16        | 96.05 ± 0.08        |
| IVGAE (ours) | ✓ | ✓ | ✓ | <b>95.66 ± 0.21</b> | <b>96.13 ± 0.14</b> |

**Table 4**

AUC and AP test results on public datasets Cora and CiteSeer. Bold and underline represent best and suboptimal, respectively.

| Method       | Cora               |                    | CiteSeer           |                    |
|--------------|--------------------|--------------------|--------------------|--------------------|
|              | AUC                | AP                 | AUC                | AP                 |
| GAE [26]     | 91.0 ± 0.02        | 92.0 ± 0.03        | 89.5 ± 0.04        | 89.9 ± 0.05        |
| VGAE [26]    | 91.4 ± 0.01        | 92.6 ± 0.01        | 90.8 ± 0.02        | 92.0 ± 0.02        |
| GNAE [25]    | <u>95.6</u>        | 95.7               | 96.5               | 97.0               |
| VGNAE [25]   | 95.4               | 95.8               | <b>97.0</b>        | 97.1               |
| MSVGAE [35]  | 95.3 ± 0.05        | 95.4 ± 0.04        | 95.4 ± 0.03        | 96.1 ± 0.04        |
| FE-GNAE [18] | 95.4               | 95.5               | <b>97.0</b>        | 97.2               |
| BW-NAE [36]  | <u>95.6</u>        | <b>96.1</b>        | 96.5               | 96.9               |
| BW-VNAE [36] | 95.3               | <u>95.8</u>        | <b>97.0</b>        | 97.0               |
| IVGAE(ours)  | <b>95.7 ± 0.21</b> | <b>96.1 ± 0.14</b> | <u>96.8 ± 0.31</u> | <b>97.3 ± 0.19</b> |

[DAGAN, GCN]+AVFF model is better than DAGAN+AVFF, indicating that the two branches designed in this article are effective. Comparing the three feature fusion methods, the AVFF technology used in this article is superior. Further, when the initial information is reinforced again by adding GCN, the prediction performance of AUC and AP improves by 0.27% and 0.14% respectively. This indicates that the initial prior information of the graph structure is crucial for the model, and it guides the network to learn better node features to obtain best prediction performance.

In addition, as shown in Table 3, the three technologies in the DAGAN module are indispensable. It can be seen that the initial information is important for extracting deep feature information of nodes in the network. The operation of normalizing the feature matrix helps the model converge faster and therefore perform better. Overall, the framework of the proposed model is reasonably designed, and each technology also plays a crucial role in improving the model's performance.

#### 4.4. Main results

To illustrate the excellence of our model, we first compare IVGAE-GA with state-of-the-art link prediction methods. As shown in Table 4, many comparative experiments are conducted on two commonly recognized benchmark datasets. Then, four representative comparative methods were selected to carry out the link prediction task in the grain import trade network. The experimental results are shown in the Tables 5 and 6. Next, we will analyze the comparative experimental results in detail.

Cora and Citeseer datasets are commonly used public datasets in link prediction tasks. These two datasets allow us to intuitively evaluate the performance of different algorithms. The proposed model in this paper is compared with some state-of-the-art methods, and the AUC and AP values of each model are summarized in Table 4. It is worth noting that the parameter settings of baseline methods vary greatly, and some papers do not provide source code. To make a fairer comparison, the accuracies of the baseline approaches in this article are taken from the original literature. Compared with these baselines, our model obtains optimal or suboptimal results in terms of AUC and AP values. This means that the six simple hyperparameters can achieve good model performance, but they cannot achieve the best results. When compared with the classic model VGAE, the performance improvement in IVGAE is significant, with 4.70% and 3.78% improvement on the Cora dataset,

**Table 5**

AUC test results on grain import trade networks. Bold and underline represent best and suboptimal, respectively.

| Method       | Soybean            | Maize              | Rice               | Wheat              |
|--------------|--------------------|--------------------|--------------------|--------------------|
| GAE [26]     | 87.0 ± 0.90        | 87.4 ± 0.72        | 86.0 ± 0.52        | 89.7 ± 0.74        |
| VGAE [26]    | 88.8 ± 0.83        | 87.2 ± 0.59        | 86.7 ± 0.66        | 88.8 ± 0.65        |
| GNAE [25]    | 92.0 ± 0.67        | <u>92.0 ± 0.47</u> | <u>90.6 ± 0.38</u> | <u>93.4 ± 0.35</u> |
| VGNAE [25]   | <u>92.6 ± 0.53</u> | 90.7 ± 0.35        | 90.3 ± 0.61        | 92.5 ± 0.37        |
| IVGAE (ours) | <b>93.5 ± 0.41</b> | <b>95.4 ± 0.23</b> | <b>93.4 ± 0.20</b> | <b>94.0 ± 0.25</b> |

**Table 6**

AP test results on grain import trade networks. Bold and underline represent best and suboptimal, respectively.

| Method       | Soybean            | Maize              | Rice               | Wheat              |
|--------------|--------------------|--------------------|--------------------|--------------------|
| GAE [26]     | 88.0 ± 0.68        | 87.4 ± 0.65        | 86.1 ± 0.46        | 88.5 ± 0.65        |
| VGAE [26]    | 90.2 ± 0.57        | 87.1 ± 0.48        | 86.8 ± 0.64        | 86.9 ± 0.51        |
| GNAE [25]    | 91.8 ± 0.57        | <u>91.6 ± 0.31</u> | 87.8 ± 0.27        | <u>93.0 ± 0.21</u> |
| VGNAE [25]   | <u>92.2 ± 0.31</u> | 90.1 ± 0.22        | <u>90.1 ± 0.49</u> | 92.1 ± 0.32        |
| IVGAE (ours) | <b>94.5 ± 0.43</b> | <b>95.8 ± 0.40</b> | <b>93.2 ± 0.23</b> | <b>94.6 ± 0.17</b> |

and 6.61% and 5.76% improvement on the Citeseer dataset. This indicates that our model successfully captures graph structure information and efficiently learns useful features of each node through iterative propagations.

As mentioned in Section 2.2, GAE and VGAE are classic models that apply graph autoencoder technology to link prediction tasks, and many researchers utilize them as the basic framework to build link prediction models. For GNAE and VGNAE models, the latent representation of nodes is obtained through the APPNP [39] module. Different from the two-layer graph convolutional neural network, the APPNP module captures high-order information and initial information through a deeper network, while taking into account isolated nodes in the graph structure. Therefore, this paper specifically selected these four representative baseline methods to verify the excellence of our model in link prediction for four real grain import trade networks. We conducted comparative experiments under the same parameter settings and experimental environment as in this article. Tables 5 and 6 show the AUC and AP values respectively. Experimental results show that the prediction performance of our model has been greatly improved on four real grain datasets. Meantime, it can be seen that the GNAE and VGNAE methods perform better than the GAE and VGAE models, which indicates that the high-order information in the graph structure is very important for link prediction. The performance improvement is attributed to the IVGAE's ability to effectively mine node feature information and more effectively balance the fusion relationship between the initial prior information and the extracted latent representation.

#### 4.5. discussion

##### 4.5.1. Analysis of computational cost

Taking the Cora dataset as an example, the proposed model is compared with some representative baseline methods regarding the number of parameters and computational complexity. As seen from Table 7, we select methods based on variational graph autoencoders for comprehensive comparison under the approximate framework. The parameters and FLOPs of the proposed IVGAE method are more numerous than other methods. The high complexity of the model is mainly because we stacked multiple DAGAN modules to alleviate drawbacks of the static graph structure, which increased complexity several times. When we only used one DAGAN module, the complexity of our model (IVGAE-one) was greatly reduced while obtaining suboptimal results. In addition, the BW-VNAE model achieves the lowest complexity, but the random walk process of this model takes a lot of training time. While computational cost is an important factor, improving the performance of the model is also particularly important. Within an

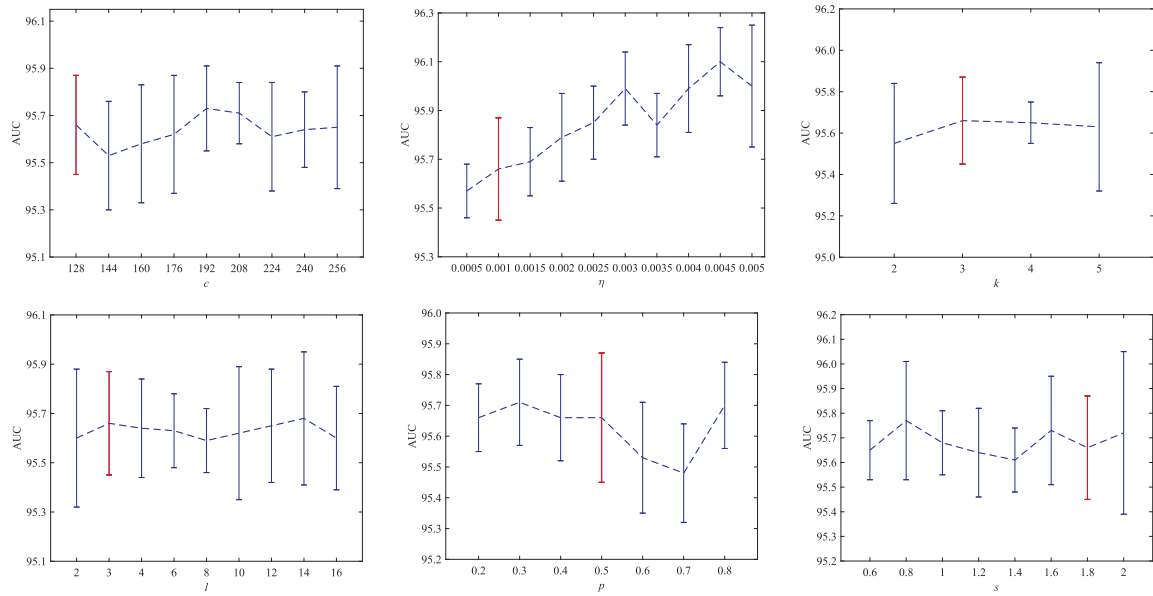


Fig. 5. Comparison of AUC test results after adjusting hyperparameter settings on Cora dataset.

Table 7

Number of parameters and FLOPs of different methods on the Cora dataset. “M” and “G” represent millions and billions, respectively.

| Method           | FLOPs (G) | Parameters (M) | AUC          | AP           |
|------------------|-----------|----------------|--------------|--------------|
| VGAE [26]        | 1.17      | 0.22           | 91.4 ± 0.01  | 92.6 ± 0.01  |
| VGNAE [25]       | 0.99      | 0.37           | 95.4         | 95.8         |
| MSVGAE [35]      | 1.99      | 0.74           | 95.3 ± 0.05  | 95.4 ± 0.04  |
| BW-VNAE [36]     | 0.13      | 0.05           | 95.3         | 95.8         |
| IVGAE-one (ours) | 0.99      | 0.37           | 95.50 ± 0.16 | 96.06 ± 0.08 |
| IVGAE (ours)     | 2.98      | 1.10           | 95.66 ± 0.21 | 96.13 ± 0.14 |

Table 8

Selection ranges of six hyperparameters.

| Dataset  | $c$     | $\eta$       | $k$ | $L$  | $p$     | $s$ |
|----------|---------|--------------|-----|------|---------|-----|
| Cora     | 128~256 | 0.0005~0.005 | 2~5 | 2~16 | 0.2~0.8 | 0~2 |
| CiteSeer | 256~512 | 0.0005~0.005 | 2~5 | 2~16 | 0.2~0.8 | 0~2 |
| Soybean  | 10~20   | 0.0005~0.005 | 2~5 | 2~16 | 0.2~0.8 | 0~2 |
| Maize    | 10~20   | 0.0005~0.005 | 2~5 | 2~16 | 0.2~0.8 | 0~2 |
| Rice     | 10~20   | 0.0005~0.005 | 2~5 | 2~16 | 0.2~0.8 | 0~2 |
| Wheat    | 10~20   | 0.0005~0.005 | 2~5 | 2~16 | 0.2~0.8 | 0~2 |

acceptable computational cost, our model performance is significantly better than other methods. Compared with other large models, the proposed method achieves a better balance between accuracy and complexity.

#### 4.5.2. Sensitivity analysis of hyperparameters

The hyperparameters in our model mainly include the output channel  $c$ , the learning rate  $\eta$ , the number of DAGAN modules  $k$ , the layer  $L$ , the dropping rate  $p$ , and the scaling constant  $s$ . The numerous parameters make it impractical to exhaustively search through all combinations. To find a good balance between efficiency and performance, we adopt sensitivity analysis to systematically identify the impact of each parameter on performance. Specifically, we take the Cora dataset as an example to conduct sensitivity analysis on six parameters and study the impact of their changes on model performance.

As depicted in Figs. 5 and 6, we achieved practical and informed hyperparameter tuning under resource constraints. By adjusting each hyperparameter individually while keeping others constant, the performance of the model exhibited fluctuations, and performance improvements were observed in each case. Among the six parameters, the

performance of our model is more sensitive to changes in the learning rate  $\eta$ , with AUC and AP values improved by 0.42% and 0.31%, respectively. Compared with other parameters, the model performance is the least sensitive to the adjustment of parameter  $k$ , and the improvement is not significant, but the training time is increased. These findings are consistent with theoretical expectations. The learning rate determines the magnitude of weight updates during the gradient descent process and is a key driver of optimization efficiency. Due to the robustness of the model architecture, there was no significant performance degradation caused by problems such as over-smoothing throughout the hyperparameter tuning process.

#### 4.5.3. Tuning all hyperparameters based on genetic algorithm

Sensitivity analysis has inherent limitations, particularly the inability to account for synergies between hyperparameters, hindering the identification of the global optimal parameter configuration. To effectively address this issue, we further use the genetic algorithm to search for the globally best hyperparameter combination in the complex hyperparameter space to obtain better link prediction results. Table 8 shows the value range of each hyperparameter when applying the genetic algorithm. Due to time cost and computer memory limitations, this article sets the initial population size to 3 and the number of iterations to 3. Specifically, three initial populations are first randomly generated within the specified value range, and the hyperparameter settings are evaluated through model training and the AP value. Then, the population with excellent performance is selected as the parent generation, and blend crossover and Gaussian mutation are applied to optimize the hyperparameter settings. Finally, through continuous iteration, selection, improvement and other processes, each hyperparameter setting can be rigorously tested until the best performing hyperparameter configuration is returned, and the whole process needs to be repeated 21 times in total. Compared with sensitivity analysis, genetic algorithms not only avoid exponential search costs, but also reduce the risk of local optimality, effectively balancing tuning and costs.

Table 9 presents the performance improvement of the model in this article after using the genetic algorithm. From the perspective of prediction performance and model stability, our method has been improved on all datasets. Comparing the experimental results in Tables 4, 5, and 6, the proposed model achieved the best model performance in all datasets after using the genetic algorithm. This shows that the setting of hyperparameters is very important to the model, and the designed



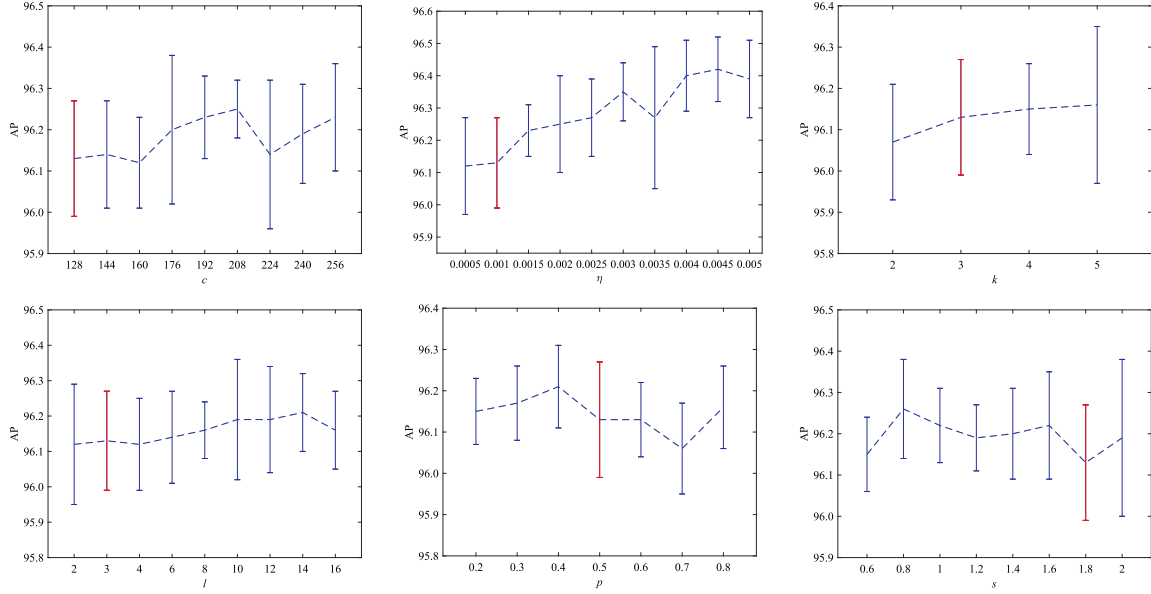


Fig. 6. Comparison of AP test results after adjusting hyperparameter settings on Cora dataset.

Table 9

Comparison of results after using the genetic algorithm on the Cora, CiteSeer, Soybean, Maize, Rice and Wheat datasets.

|     | Method         | Cora                              | CiteSeer                          | Soybean                           | Maize                             | Rice                              | Wheat                             |
|-----|----------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| AUC | IVGAE(ours)    | 95.7 $\pm$ 0.21                   | 96.8 $\pm$ 0.31                   | 93.5 $\pm$ 0.41                   | 95.4 $\pm$ 0.23                   | 93.4 $\pm$ 0.20                   | 94.0 $\pm$ 0.25                   |
|     | IVGAE-GA(ours) | <b>96.1 <math>\pm</math> 0.13</b> | <b>97.0 <math>\pm</math> 0.15</b> | <b>94.0 <math>\pm</math> 0.45</b> | <b>95.8 <math>\pm</math> 0.27</b> | <b>93.6 <math>\pm</math> 0.21</b> | <b>95.5 <math>\pm</math> 0.18</b> |
| AP  | IVGAE(ours)    | 96.1 $\pm$ 0.14                   | 97.3 $\pm$ 0.19                   | 94.5 $\pm$ 0.43                   | 95.8 $\pm$ 0.40                   | 93.2 $\pm$ 0.23                   | 94.6 $\pm$ 0.17                   |
|     | IVGAE-GA(ours) | <b>96.4 <math>\pm</math> 0.10</b> | <b>97.3 <math>\pm</math> 0.11</b> | <b>95.0 <math>\pm</math> 0.37</b> | <b>96.1 <math>\pm</math> 0.15</b> | <b>93.7 <math>\pm</math> 0.16</b> | <b>96.0 <math>\pm</math> 0.12</b> |

model is excellent. In addition, some unsatisfactory results (such as the AP results of the CiteSeer dataset) are attributed to inappropriate parameter settings in the genetic algorithm, and the optimal results could not be found in limited experiments.

## 5. Conclusion

In this paper, we develop a deep graph neural network for link prediction tasks, termed IVGAE-GA. The entire architecture improves the feature extraction method under the framework of the VGAE. The fine feature representation of nodes comes from two parts: non-local feature information and local feature information. Among them, non-local features containing high-order information are mainly captured through multiple DAGAN modules. Then, we use an average fusion operation to blend the feature information of different node views. GCN is used to obtain local information containing domain feature information. Extensive comparison and ablation experiments demonstrate that the proposed IVGAE-GA method has a reasonable framework and superior performance compared with some existing recommendation methods.

The link prediction method proposed in this paper can well predict the potential paths and future trends of grain trade networks. The findings from this study provide insights for countries to ensure food security and promote sustainable food trade. Compared with traditional network methods, graph neural networks applied in this paper have achieved better experimental results in the link prediction task. This is accomplished through mining the initial prior information and graph structure information from the trade network. This work expands the application of graph neural networks in the field of supply chain risk and enriches the understanding of risk management in supply chain security. It provides powerful tools for countries to solve practical

problems such as risk assessment and early warning of import supply chains. Meanwhile, there remains potential for further research about this work. From the perspective of graph structure, we only focus on static graphs. This structural assumption limits the model's ability to capture the dynamic evolution of real trade networks. Future research can encode each trade relationship in dynamic trade networks to mitigate this deficiency by constructing link prediction models based on dynamic graphs. In addition, advanced algorithms should be further designed for link prediction tasks on large-scale graph data.

## CRedit authorship contribution statement

**Yanhui Li:** Writing – review & editing, Supervision, Methodology, Investigation. **Yuzhi Song:** Writing – original draft, Software, Methodology, Data curation. **Qi Yao:** Supervision, Resources, Funding acquisition. **Xu Guan:** Writing – review & editing, Supervision, Funding acquisition.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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